

# The Architecture of Modern LLMs

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## 1 Introduction, History, Motivation

The LLM is the foundation for AI and has been in development for almost two decades, but has only become mainstream in the last couple of years. One of these changes that contributed to societies big pivot into AI was the introduction of the "transformer", changing the algorithm used in LLMs, and restructuring how they work. This report goes into depth specifically on the differences between past and present day LLM architectures, and how more efficient algorithms led to the exponential growth of AI.

Before 2017, LLMs before GPT-3 relied on recurring neural networks (RNNs) and LSTMs, which sequentially processed text. This caused a couple of issues as AI endured bigger work loads and more complex tasks. If you visualize a game of telephone with an extremely long line of people, and started with a complex word, the person at the end of the line would have an entirely different word. This is what would happen to old LLM tasks, as they faced hallucination, syntax loss, and just severe bottleneck in general.

The Transformer architecture solves this by using self-attention to process all words in a sequence simultaneously. Each word is represented as a vector transformed into Query ( $Q$ ), Key ( $K$ ), and Value ( $V$ ) components. The attention scores are calculated as:

$$\text{Attention}(Q, K, V) = \text{softmax} \left( \frac{QK^T}{\sqrt{d_k}} \right) V$$

This mechanism allows the model to disambiguate meaning based on the entire context of a sentence. Building on this, the GPT-3 research illustrates that scaling this architecture leads to powerful emergent behaviors, allowing for zero-shot and few-shot learning.

I chose this topic because of the impact it has on society. Millions of people are utilizing AI every hour, and yet 90 percent of users don't even know how it works. As students entering this technology industry, we use AI for every task. This is what sparked my interest in understanding the fundamentals of AI; if I was going to be using this tool for the rest of my career, it is crucial for me to get a better grasp of how it works, in order to develop my career further.

We did run into issues of course, like any project. With the AI industry being super large, it is impossible to constrain all the relevant information into a single 20 minute lecture. Along with that, there are so many different topics, it felt impossible to simply pick one. But I decided on specifically picking the algorithm on how AI reads text. I chose this in particular

because it is the most common use function for majority of AI users. People will use AI to scan tasks, documents, books, etc; so focusing on that specific part of AI felt appropriate, as it would better transition into how its new algorithm, the Transformer, would go on to impact the world.

## **2 Conclusion**

### **2.1 Drawbacks**

One drawback we had was struggling to balance all of our points on the main topic / arguments within in the allocated time, while also trying to provide all the relevant context on history and the industry. Between my partner and I, there was so many points that we wanted to bring up to further strengthen our cases and lecture, but we ended up having to scrap it. By only focusing on key components, such as the different between RNN and Transformer Algorithms, as well as the impact on society, we were still able to structure the lecture in a way that explain the topic well.

### **2.2 Future Work/Applications**

If we wanted to expand on this, the first thing I would do would be to go back and actually include a ton of the relevant historical context that was previously left out. By adding that, it sets up the foundations for other algorithms / points to be studied. A good example would be how the algorithms for Photo detections and Generation. Another point that we would love to add / introduce is how the concept of matrix's were used in the first place, as this is the ultimate starting point of AI. It would grant us a better foundation to further explain math concepts and provide better dissection of present and future topics.

### **2.3 Summary**

Our findings conclude that the catalyst for the massive change in society was the transition from RNNs algorithms to the transformer algorithm. It was challenging to pinpoint our argument just to this for the sake of the lecture's constraints, but I believe it was well executed. Something we could have done differently was create a more interactive activity for everyone during the lecture, but we struggled with making an activity that would be fun and educational, without confusing everyone.

## **3 Contributions**

### **3.1 Review on Myself**

I believe that I lectured my part of the presentation very well. My points broke dow the topic well and the information was digestible. I had clear graphical and examples that further strengthened by lecture, and I genuinely believe that I carried my end of the presentation to my fullest extent. I believe I deserve an A.

### 3.2 Review on my Partner

My partner, Ahmed, was an absolute pleasure to work with. We saw each other daily so communication was not an issue. He was cooperative and supportive of ideas, and contributed a decent amount. My only critique would be during the actual lecture itself, as he isn't super confident in his ability to publicly speak, but I did not find it an issue at all. I believe he also deserves near an A.

### 3.3 Work Split

| Project Part            | Who did it?      | More Details                                                                                                                          |
|-------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Initial Proposal        | Shahid and Ahmed | I proposed the idea on AI's algorithms, and he proposed to scale the concepts to global influence.                                    |
| Created Examples        | Shahid           | Created the examples on the differences between RNNs and Transformers.                                                                |
| Topic A: The Algorithms | Shahid           | Researched past and present algorithms, and explained the exact formulas on how the Transformer component works in the new algorithm. |
| Topic B: The Scope      | Ahmed            | Researched how the Transformer was able to scale up AI usage and functionality into its present-day impact.                           |

Table 1: Project Breakdown

## References

- [1] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30.
- [2] Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., ... & Amodei, D. (2020). Language models are few-shot learners. *arXiv preprint arXiv:2005.14165*.